

Assignment 1: Unsupervised Learning Methods (group 5)

Clustering Patient Data: A Comparative Analysis

1. Data Loading & Exploration

1.1. Data loading and initial exploration

We loaded the Heart Disease dataset (with 303 patients and 13 features) using the *ucimlrepo* package.

Feature	Name	Description
age	Age	Age in years
sex	Sex	1 = male, 0 = female
cp	Chest pain type	4 types (0–3)
trestbps	Resting blood pressure	mm Hg on admission
chol	Serum cholesterol	mg/dl
fbs	Fasting blood sugar	>120 mg/dl: 1 = true, 0 = false
restecg	Resting ECG results	Values 0, 1, 2
thalach	Max heart rate achieved	Numeric
exang	Exercise-induced angina	1 = yes, 0 = no
oldpeak	ST depression	Induced by exercise vs. rest
slope	Slope of peak ST segment	Values 0, 1, 2
ca	Major vessels (0–3)	Coloured by fluoroscopy
thal	Thal	3 = normal, 6 = fixed defect, 7 = reversible defect
target	Diagnosis (HIDDEN)	0 = no disease, 1–4 = disease present. Binarise as 0 vs. 1+

Figure 1. Overview of dataset features

A summary of all variables in the Heart Disease dataset, including their names and descriptions. This provides context for the clustering analysis and helps interpret the results.

When looking at the summary statistics, we noticed that the features have different types of values and magnitudes (for example, *chol* goes up to 564, while *oldpeak* only reaches 6.2). We will use clustering algorithms that use distances, so we will need to standardize or normalize our dataset.

1.2. Missing values

Now we checked for missing values and found only 6 in total (4 in *ca* and 2 in *thal*). We could use the median as this is only a 1.9% of NAs, but we wanted a more robust approach. We assumed the data is MAR (missing at random) and we used a KNN imputer with $k=5$ to fill the gaps based on similar patients. As KNN relies on distances, we did a temporary standardization.

1.3. Plotting features

Finally, we plotted histograms and boxplots. The histograms showed that our features have very different distributions (normal, skewed, categorical,...). The boxplots showed the outliers on variables like cholesterol and resting blood pressure.

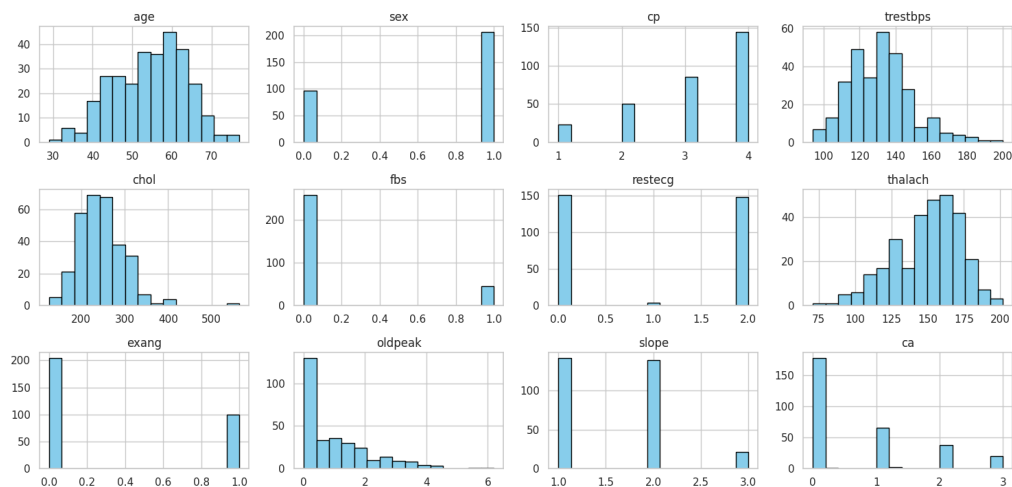


Figure 2. Histograms of all features in the Heart Disease dataset, showing differences in distributions across variables, including normal, skewed, and categorical patterns.

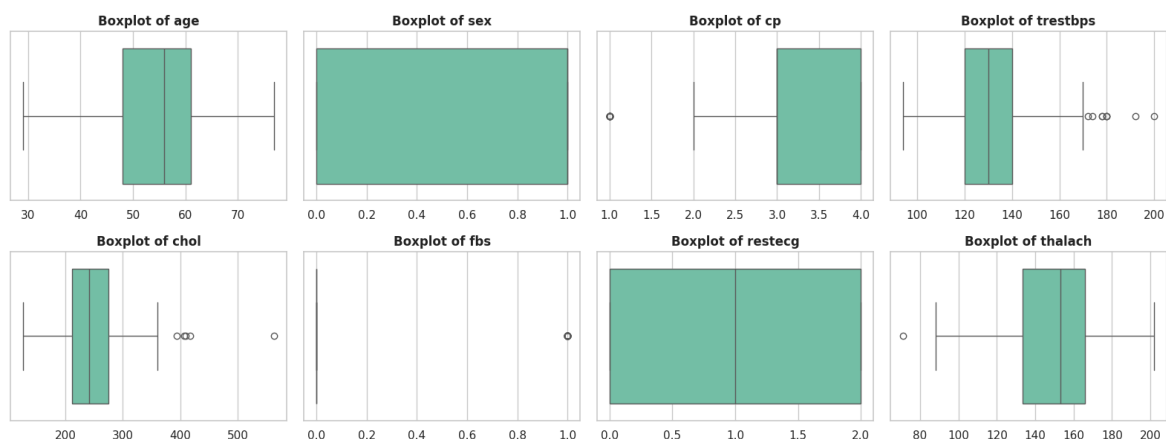


Figure 3. Boxplots of all features highlighting outliers, particularly in variables such as *trestbps* or *chol*.

Correlation heatmap

To ensure no redundant information we generated a correlation heatmap. There was no significant correlation between features (maximum of 0.58 between *oldpeak* and *slope*). This shows that any feature is trivially redundant, so we cannot reduce the dimensionality and a PCA is needed.

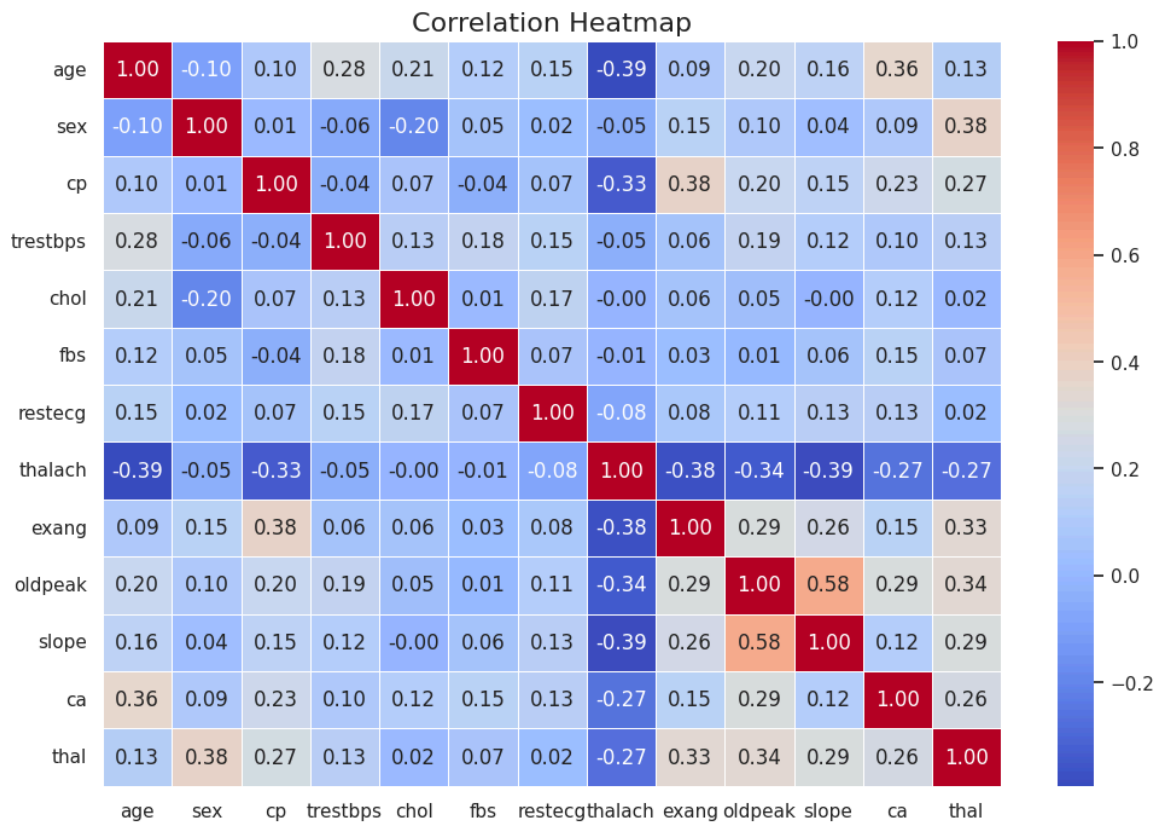


Figure 4. Correlation heatmap

2. Preprocessing & Feature Engineering

In order to prepare the dataset for clustering, several preprocessing steps were applied, including categorical encoding, feature scaling, dimensionality reduction using Principal Component Analysis (PCA), and supervised feature selection. These steps ensure that the data is suitable for distance-based clustering algorithms and that irrelevant or redundant features are minimized.

2.1 Encoding of Categorical Variables

Several variables in the dataset represent categorical information, including chest pain type (`cp`), resting ECG results (`restecg`), slope of the ST segment (`slope`), and thalassemia (`thal`). These features were transformed using one-hot encoding.

One-hot encoding was chosen over label encoding because it avoids introducing artificial ordinal relationships between categories. Since clustering algorithms rely on distance calculations, treating categorical values as ordered integers could distort similarity measures. By converting each category into a binary feature, the model can treat them independently and more accurately reflect their contribution to patient similarity.

To prevent redundancy and multicollinearity, the first category of each variable was dropped (`drop_first=True`).

2.2 Feature Scaling

All features were standardised using the `StandardScaler`, which transforms each feature to have zero mean and unit variance.

This step is essential because both PCA and clustering algorithms such as K-Means are sensitive to the scale of input features. Without scaling, variables with larger numerical ranges (e.g., cholesterol or blood pressure) would dominate distance calculations, leading to biased clustering results. Standardisation ensures that all features contribute equally to the analysis.

To avoid data leakage, the dataset was split into training and test sets prior to scaling. The scaler was fitted only on the training data and then applied to both training and test sets.

2.3 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to the scaled dataset to analyse the variance structure and reduce dimensionality for visualisation purposes.

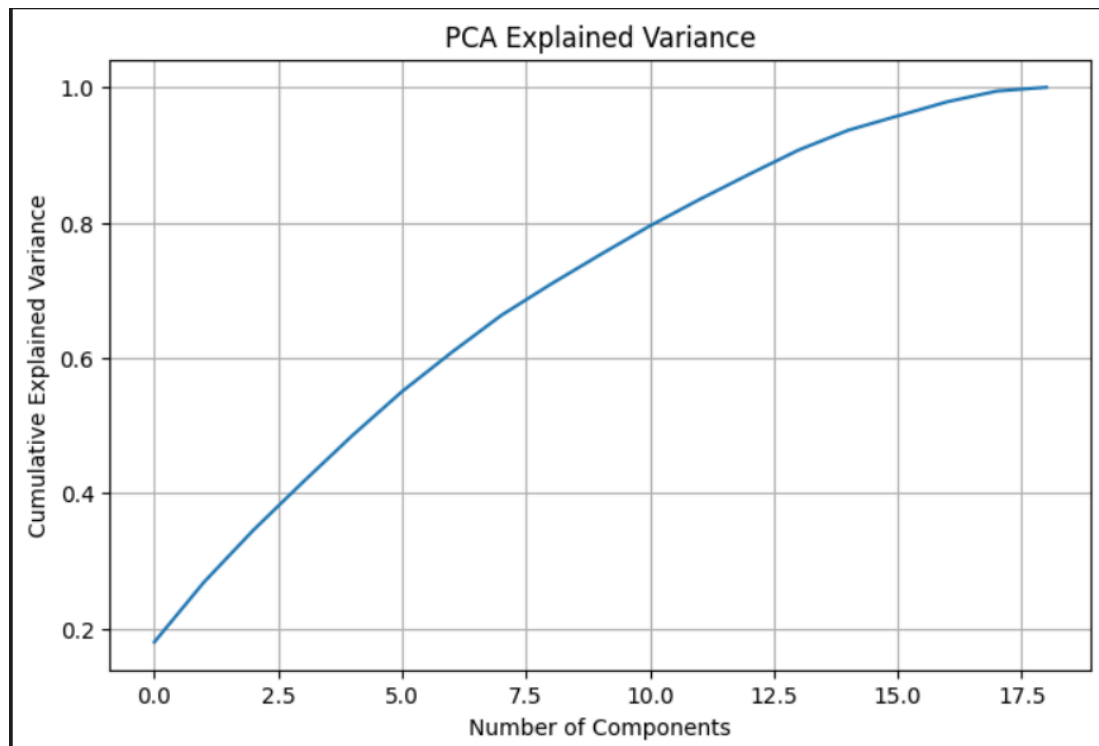


Figure 5. Explained variance ratio of each principal component, showing how much variance is captured as the number of components increases.

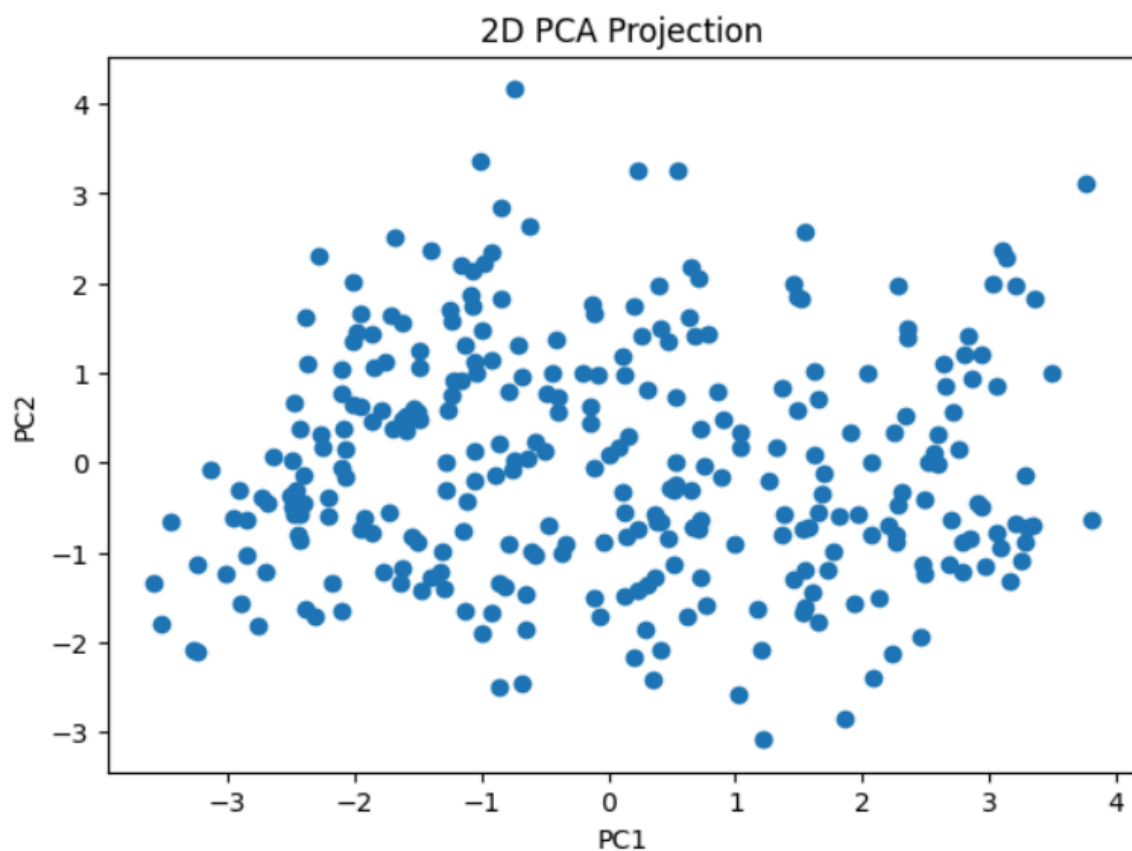


Figure 6. Projection of the dataset onto the first two principal components, illustrating the distribution of this variable in reduced-dimensional space.

The cumulative explained variance plot indicated that approximately 13 principal components were required to retain at least 85% of the total variance. This suggests that the dataset has moderate intrinsic dimensionality, with most features contributing meaningful information and limited redundancy.

A two-dimensional PCA projection was also generated to visualise the data. The resulting scatter plot did not show clearly separable clusters, indicating that the underlying structure of the dataset may not be strongly clustered in low-dimensional space. This suggests that clustering algorithms may face challenges in identifying well-defined patient groups.

It is important to note that PCA was used only for exploratory analysis and visualisation, and not as input for clustering.

2.4 Feature Selection

Supervised feature selection was performed using the SelectKBest method with the ANOVA F-test. This is the only step in the pipeline where the target variable was used, as permitted by the assignment.

The method evaluates the statistical relationship between each feature and the target variable, selecting those with the highest discriminatory power. A subset of the top 8 features was selected to balance dimensionality reduction with information retention.

The selected features included variables such as maximum heart rate (**thalach**), exercise-induced angina (**exang**), ST depression (**oldpeak**), and number of major vessels (**ca**). These features are clinically relevant indicators of heart disease, supporting the validity of the selection process.

After feature selection, the dataset was reduced to 8 features and re-scaled to ensure consistency for clustering.

2.5 Comparison of PCA and Feature Selection

PCA and supervised feature selection capture different aspects of the dataset.

PCA identifies directions of maximum variance without considering the target variable, resulting in linear combinations of features that are often difficult to interpret. In contrast, feature selection methods such as SelectKBest identify features that are most strongly associated with the target variable, making the results more interpretable and clinically meaningful.

While PCA suggested that many components are required to capture the variance in the data, feature selection reduced the dataset to a smaller subset of highly informative variables. This indicates that not all variance in the dataset is relevant for distinguishing between patients with and without heart disease.

2.6 Final Feature Set

The final dataset used for clustering consists of 303 samples and 8 selected features, all standardised. This representation balances dimensionality reduction with information preservation and provides a suitable input for the clustering algorithms applied in subsequent tasks.

3. K - Means Clustering

The K-means analysis identified two clusters in the dataset: Cluster 0 included 121 patients (39.9%), whereas Cluster 1 included 182 patients (60.1%).

The PCA visualisation showed some overlap between the two groups, suggesting that the separation was present but not particularly strong. In the original feature space, Cluster 0 was characterised by lower thalach and higher values of exang, oldpeak, ca, cp_4.0, slope_2.0, and thal_7.0, while Cluster 1 showed the opposite pattern, with higher thalach and cp_3.0.

Overall, the results indicate that K-means was able to distinguish two patient profiles, although the boundary between the clusters remained moderate rather than clearly defined.

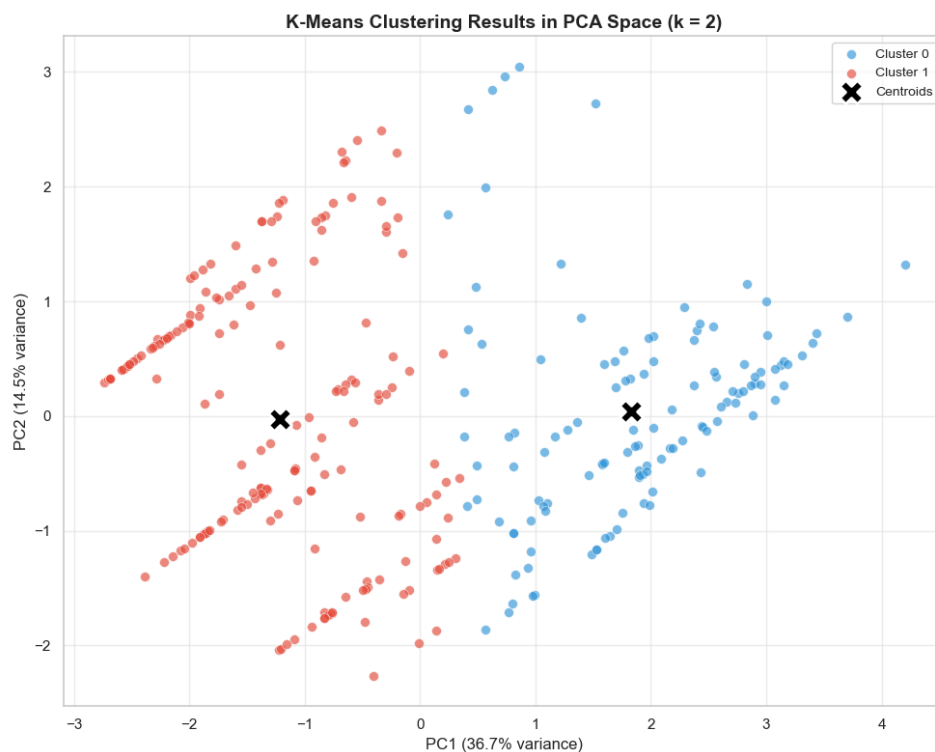


Figure 7. PCA projection of the K-means clustering result (k=2).

The two groups are visible, but they still partially overlap, which is consistent with a moderate separation between clusters.

4. Gaussian Mixture Models Clustering

The Gaussian Mixture Model analysis identified four clusters in the dataset, each containing a similar proportion of patients (approximately 21–28%).

The code for this analysis is available at: <https://github.com/harasarahh/machine-learning-projects.git>

The PCA visualisation showed a clearer structure than K-Means, with clusters forming elongated, roughly parallel ellipses and only limited overlap between groups.

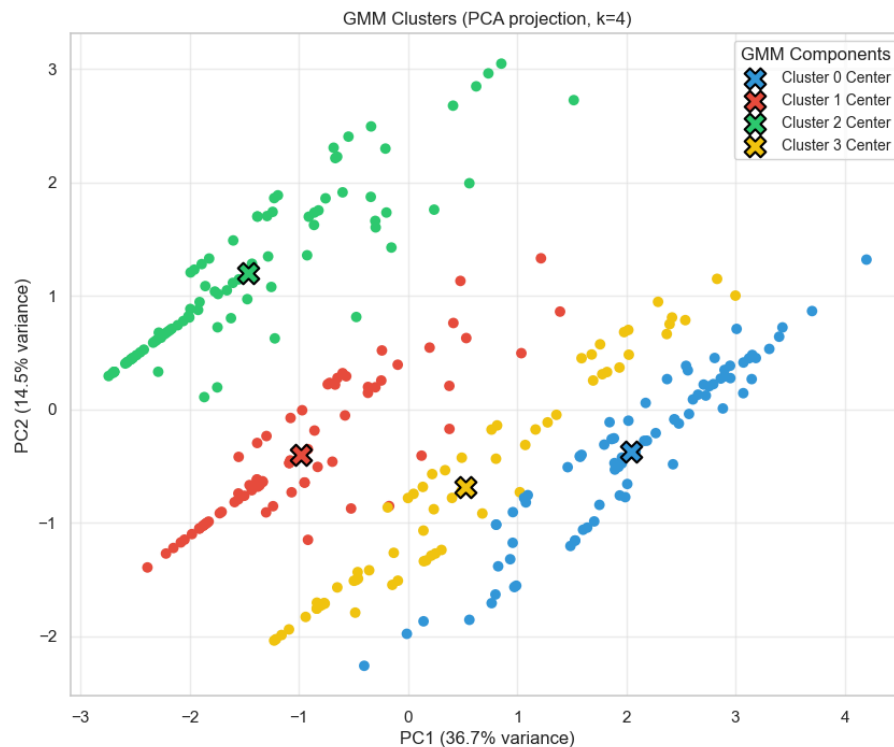


Figure 8. PCA projection of the GMM clustering result (k=4).

Most points are clearly grouped, with minimal overlap between clusters. The four GMM clusters form four elongated, parallel ellipses, with their major axes aligned along a direction with an approximate slope of 1.

In the original feature space, the clusters were mainly differentiated by chest pain type (cp_*), exercise-induced angina ($exang$), and maximum heart rate ($thalach$), while features such as $oldpeak$, $thal_7.0$, and ca showed similar patterns across clusters. Cluster 0 was associated with higher-risk indicators (e.g., $exang$ and $cp_4.0$), whereas the remaining clusters displayed different combinations of $thalach$ and chest pain-related features.

Overall, the results indicate that GMM captures a more detailed structure of the dataset, identifying four distinct clusters with different characteristics compared to the two broad groups found by K-Means.

5. Hierarchical Clustering

For this part, we used **Ward's method** to group the data. The analysis divided the patients into two almost equal groups: **Cluster 0** with 154 patients (51%) and **Cluster 1** with 149 (49%).

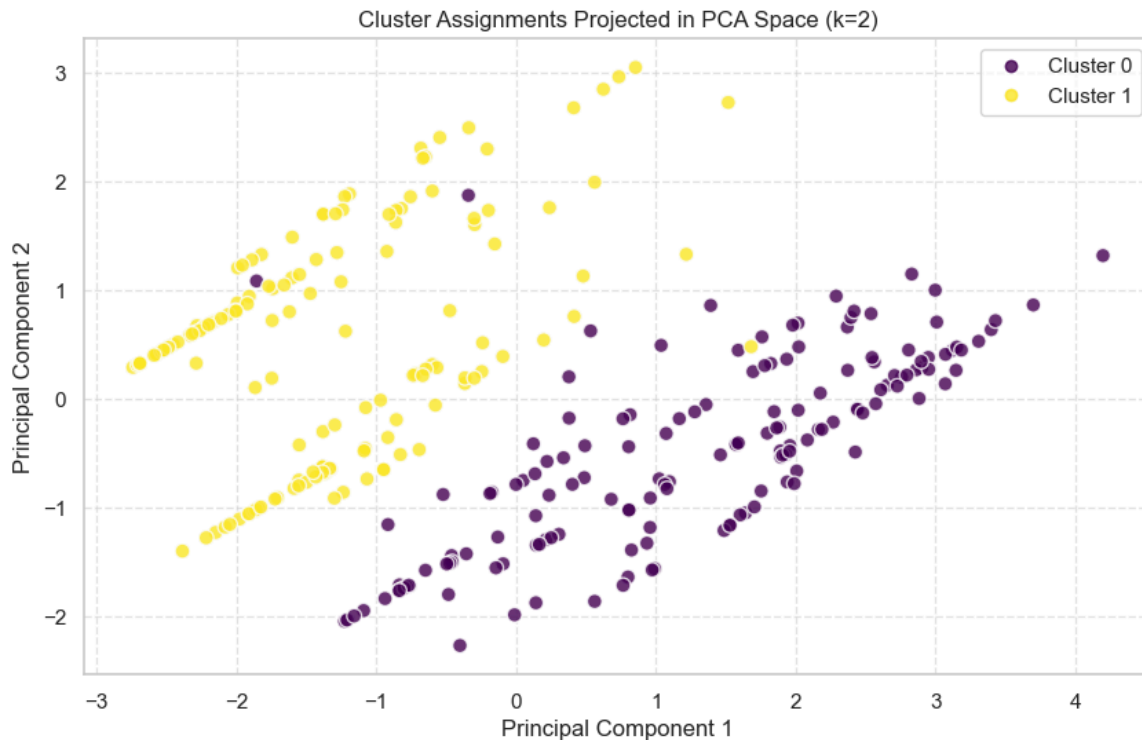


Figure 9. PCA projection of the Hierarchical clustering result ($k=2$).

The PCA visualization showed a fairly clear separation between the two groups. The main differences we found in the features were:

- **Cluster 0:** This group includes patients with higher risk indicators, such as *oldpeak*, *ca* (number of major vessels), and *exang* (exercise-induced angina). This suggests it represents the higher-risk profile.
- **Cluster 1:** This group is characterized by a higher *thalach* (maximum heart rate) and fewer diagnostic abnormalities.

6. Method Comparison

The three clustering methods highlighted different aspects of the dataset and differed in their assumptions, strengths, and limitations.

K-means assumes compact, centroid-based clusters and provides the most interpretable solution, making it effective for capturing the dominant two-group structure of the data, although it is less suitable when clusters are overlapping or not approximately spherical. In our analysis, this method showed the best agreement with the binary ground truth, suggesting that the main structure of the dataset is closer to a broad two-group separation.

GMM is more flexible because it can model clusters with different shapes and orientations, which is useful when the data may contain more complex patterns. In our results, it identified a more detailed four-cluster structure than K-means, but this solution did not align as closely with the binary ground truth and required a more difficult model selection process.

Hierarchical clustering, using Ward's method, produced two broader patient groups and offered a useful view of how observations can be grouped step by step. Its main strength is that it reveals broader structural relationships without relying on centroid-based assignment in the same way as K-means, but its limitation is that early merging decisions influence the final cluster structure and cannot be revised later.

Overall, the comparison suggests that some cluster structure is present in the data, but that the methods capture it at different levels of granularity.

7. Evaluation, Reflection and Limitation

Based on the ARI scores and the mapping accuracies, K-Means definitively best recovered the true patient groups. It achieved the highest ARI (0.4294) and a mapping accuracy of 82.84%, clearly outperforming both GMM (Accuracy: 75.91%, ARI: 0.1622) and Hierarchical Clustering (Accuracy: 73.93%, ARI: 0.2265).

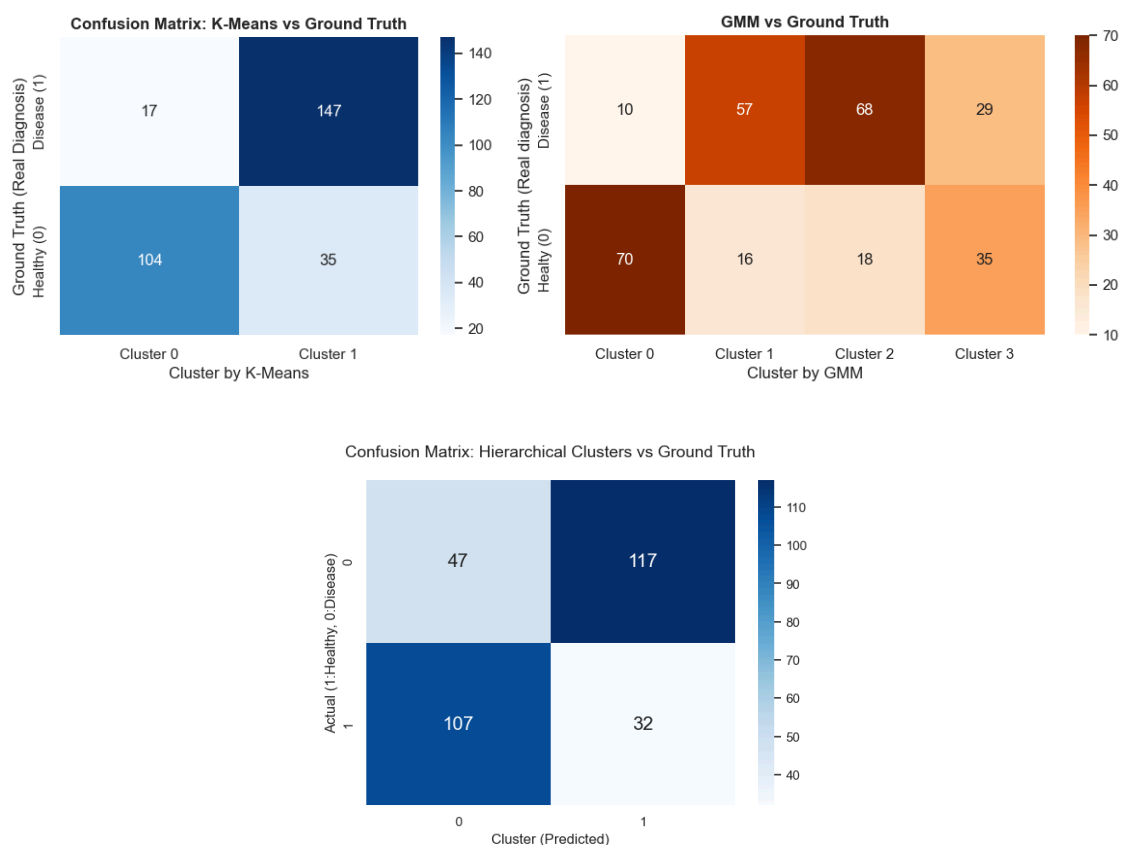


Figure 10. Confusion matrices comparing predicted clusters versus ground truth labels for K-Means (top left), GMM (top right), and Hierarchical Clustering (bottom).

It is important to note that clinical variables overlap. A healthy patient and a patient with early stage heart disease might share similar resting blood pressure or cholesterol levels. Because of this clinical spectrum, there are no perfectly separated groups in the mathematical space. Given this overlap, an accuracy of ~83% is a strong result for unsupervised learning.

Regarding the assumptions of each algorithm, the spherical clusters of the K-Means based on geometric distance effectively captured the two groups. However, GMM allows elliptical clusters gaining flexibility and was able to identify the 4 stages of disease. Consequently, when it was forced to map these 4 groups into binary ground truth, its performance dropped. The Hierarchical approach was also less accurate, maybe due his greedy approach.

Limitation

A key limitation of the analysis is the difficulty in selecting the optimal number of components k in the GMM clustering based solely on standard evaluation metrics. Both AIC and BIC decrease as k increases, suggesting improved fit but also indicating potential overfitting. For this reason, we also computed the log-likelihood and the silhouette score using hard assignments. However, similarly, the log-likelihood increases with k , while the silhouette score, although highest at $k = 2$ (which corresponds to a worse solution according to AIC and BIC), shows fluctuations across different values.

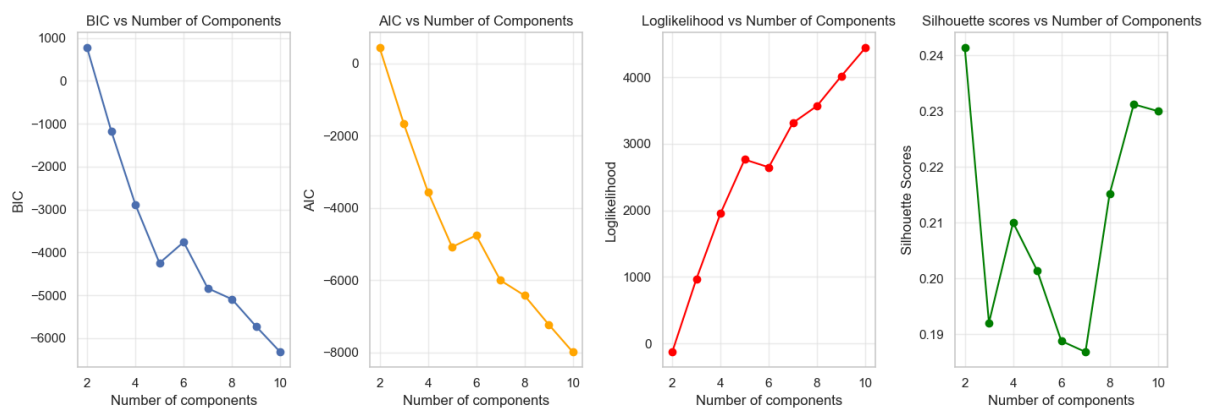


Figure 11. Evaluation of GMM clustering for different numbers of components (k). Shown are AIC, BIC, log-likelihood, and silhouette scores. No single metric provides a clear optimum, illustrating the difficulty of selecting k based solely on quantitative criteria.

As a result, the selection of $k = 4$ relied partially on visual inspection of the PCA projection, where clusters appeared more clearly separated with fewer overlaps. However, this introduces a limitation, as the PCA visualization only represents a 2D projection of the data and does not capture the full variance of the feature space.

Therefore, the chosen number of components in GMM clustering should be interpreted with caution, as it depends on a combination of quantitative metrics and subjective visual assessment rather than a single definitive criterion.

With more time, we would test the stability of the GMM solution under repeated resampling and alternative model selection criteria.